Microsoft Azure

# EDSP Mentoring Program

## Data Preparation, EDA, and Feature Engineering

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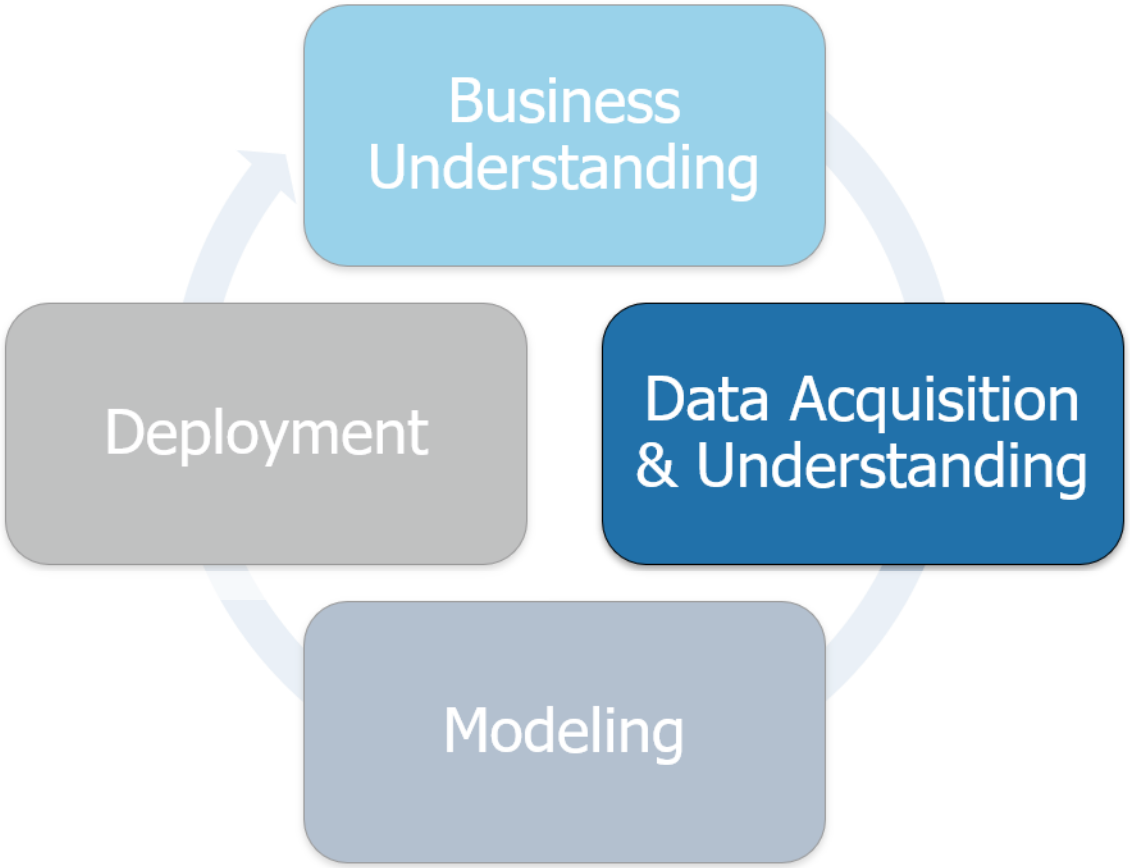
# The Microsoft Team Data Science Process

## Iterative & Exploratory: Largely Based on Conducting Experiments




**Business Understanding**
- Identify the Problem Domain
- Identify the Solution Scenario

**Acquire & Understand Data**
- Load, Prepare & Explore Data
- Identify Influential Features

**Develop Machine Learning Models**
- Select & Engineer Features
- Train, Evaluate & Tune Models

**Deployment**
- Publish Models as Webservices
- Consume Models Visually and Programmatically

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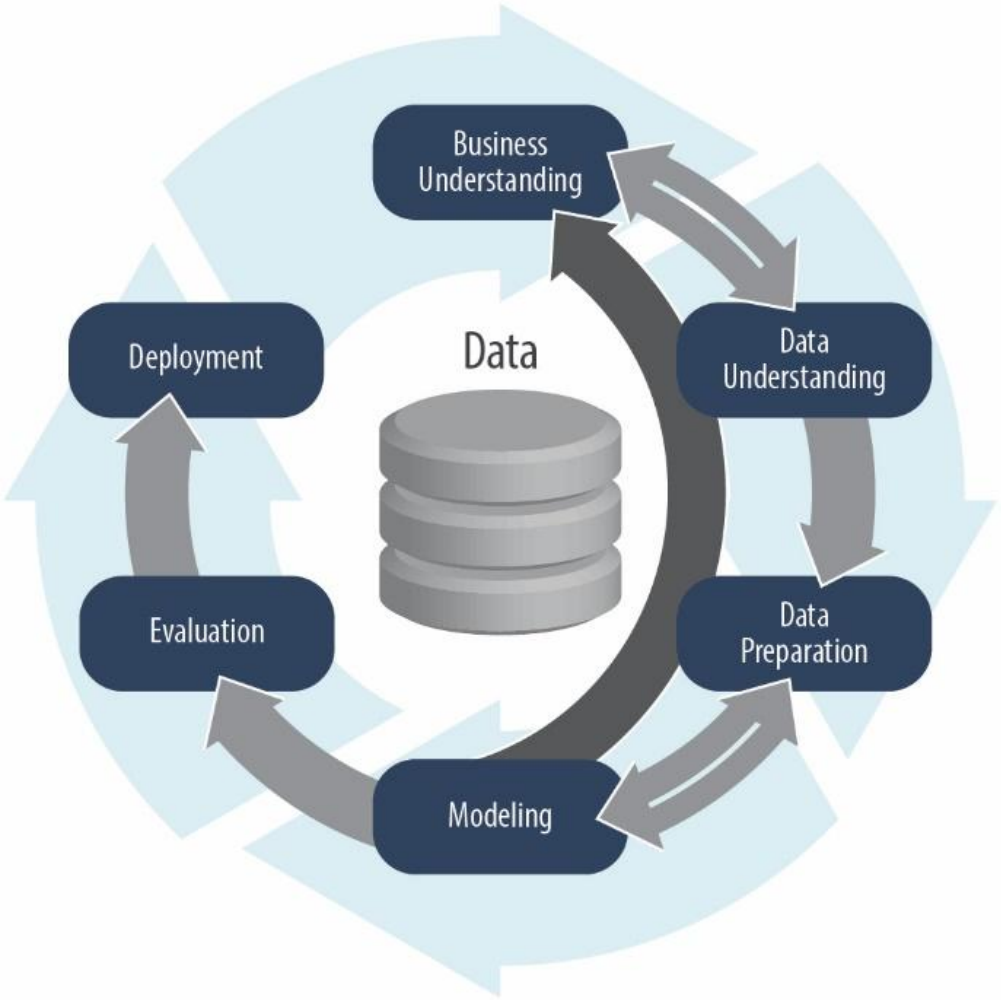
# CRISP-DM: Cross-Industry Standard Process-Data Mining

First Introduced in 1996!




**Business Understanding**
- Identify the Problem Domain
- Identify the Solution Scenario

**Data Understanding**
- Load and Explore Data
- Identify Influential Features

**Data Preparation**
- Remove Duplicates & Nulls
- Impute Missing Values
- Select & Engineer Features

**Modeling**
- Train Models Using a Variety of Algorithms
- Tune Hyper-parameters

**Evaluation**
- Test Models' Performance & Predictive Power
- Cross-Validate to Appraise Goodness-of-Fit
- Select Most Effective Model for Deployment

**Deployment**
- Publish Models On-premises or in the Cloud
- Consume Models Visually & Programmatically

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# Data Profiling: Understanding the Data

Sourcing and Understanding Data that Pertains to the Business Case

## **Identify** the *Shape* of the Dataset:

- Number of Observations (*Rows*) and their Meaning
- Number of Features (*Columns*) and their Meaning
  - DataType of Each Feature (*e.g., Numerical or Categorical*)
  - Cardinality (*Number of Unique Values*) and Rare Categories
  - Distribution of Values (*e.g., Normal, Bernoulli, Binomial, Poisson, Exponential)*
    - Statistics: *Mean, Standard Deviation, Min, Max, Interquartile Range (IQR)*

## **Identify** any *Flaws* in the Data:

- Duplicate Observations *or* Redundant Features
- Missing Data – NULL or NaN (*Not a Number*)
- Inappropriate Data Type Definitions
- Outliers – *Are they erroneous or do they represent the general truth?*

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# Data Understanding: Define the Unit of Analysis

What Does Each Row (Observation) Represent?

## Loan Risk Data

| Education | Married | Purpose | Late Payments | Annual Income | Loan Is Bad |
|---|---|---|---|---|---|
| 1 | Yes | Car Purchase | 0 | $107,000 | 0 |
| 3 | No | Small Business | 3 | $99,000 | 1 |
| 1 | Yes | House Purchase | 5 | $85,000 | 1 |
| 2 | No | Marriage | 1 | $72,000 | 0 |
| 2 | No | Debt Consolidation | 0 | $120,000 | 0 |

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# Data Understanding: Define the Target

What Do You Want to Predict?

## Loan Risk Data

| Education | Married | Purpose | Late Payments | Annual Income | Loan Is Bad |
|---|---|---|---|---|---|
| 1 | Yes | Car Purchase | 0 | $107,000 | 0 |
| 3 | No | Small Business | 3 | $99,000 | 1 |
| 1 | Yes | House Purchase | 5 | $85,000 | 1 |
| 2 | No | Marriage | 1 | $72,000 | 0 |
| 2 | No | Debt Consolidation | 0 | $120,000 | 0 |

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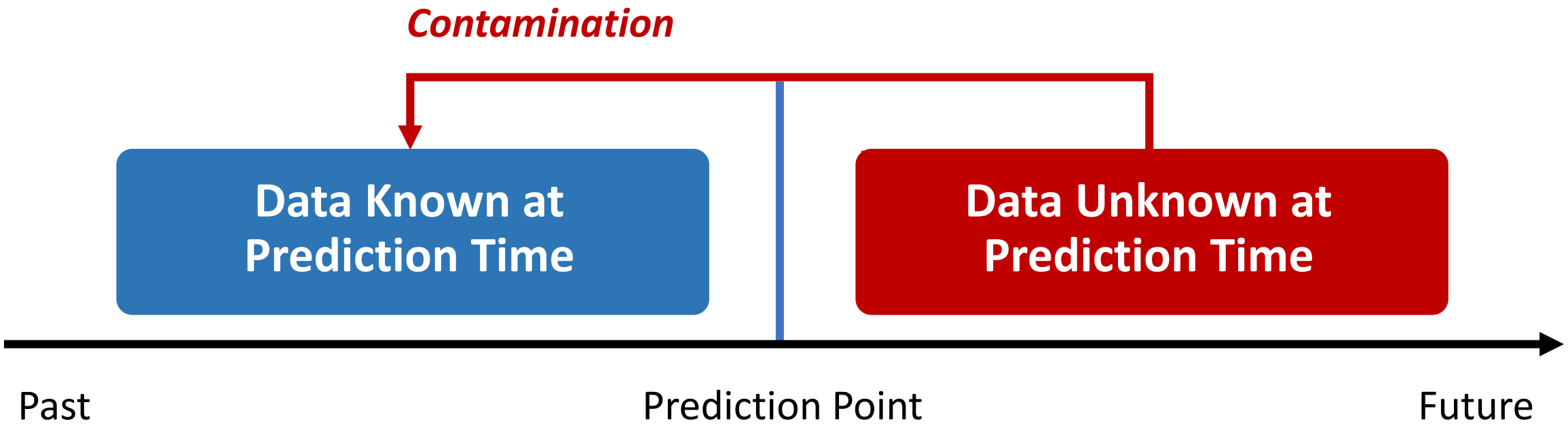
# Data Understanding: Target Leakage

Data Not Known at the Time of Prediction

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# Data Understanding: Target Leakage

Data Not Known at the Time of Prediction

## Loan Risk Data

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# Prepare Data, Engineer & Select Features

## Data Science & Machine Learning Development

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# Data Preparation: Feature Engineering

- Impute Missing Values
- Date/Time Parsing
- Scaling
- Encoding
- Generalization
- Discretization (Binning)
- Credibility Estimates

| DateOfSale |
|---|
| 03/07/2020 |
| 06/07/2020 |
| 09/02/2020 |
| 10/08/2020 |
| 12/07/2020 |

| DayOfWeek | MonthOfYear | IsWeekend |
|---|---|---|
| Saturday | March | 1 |
| Sunday | June | 1 |
| Wednesday | September | 0 |
| Thursday | October | 0 |
| Monday | December | 0 |

Age

Age Group

| DateOfBirth |
|---|
| 02/13/1969 |
| 03/05/1972 |
| 04/11/1984 |
| 05/21/1995 |
| 06/24/2002 |

| Age | Age Group |
|---|---|
| 51 | Fifties |
| 48 | Forties |
| 36 | Thirties |
| 25 | Twenties |
| 18 | Teens |

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# Feature Engineering: Numerical Features

- **Impute** missing values and create a flag to indicate imputed values:
  - Mean, Median, Mode
  - Custom or Calculated (e.g., MICE)

| Price | Mean | Median | Imputed Flag |
|---|---|---|---|
| 7 | 7 | 7 | 0 |
| null | 7 | 5 | 1 |
| 5 | 5 | 5 | 0 |
| NaN | 7 | 5 | 1 |
| 9 | 9 | 9 | 0 |

- **Scaling**:
  - **Standardize**: Rescale so mean (μ) = 0 and Standard Deviation (σ) = 1

$$z = (x_i - \mu)/\sigma$$

  - **Normalize**: Rescale so the range falls between 0 and 1

$$z = (x - \min(x)) / (\max(x) - \min(x))$$

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# Feature Engineering: Encoding Categorical Features

*Because machine learning algorithms cannot interpret text data, categorical features must first be transformed (encoded) into numerical values.*

## One-Hot Encoding

| Group | Group A | Group B | Group C |
|---|---|---|---|
| A | 1 | 0 | 0 |
| B | 0 | 1 | 0 |
| C | 0 | 0 | 1 |
| A | 1 | 0 | 0 |
| A | 1 | 0 | 0 |

*aka, Dummy Features*

## Count Encoding

| Group | Count |
|---|---|
| A | 3 |
| B | 1 |
| C | 1 |
| A | 3 |
| A | 3 |

## Ordinal Encoding

| Group | Ordinal |
|---|---|
| A | 0 |
| B | 1 |
| C | 2 |
| A | 0 |
| A | 0 |

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# Feature Engineering: Credibility Estimates

*Categorical features often have unequal member distributions.*
*Credibility estimates help compensate for this imbalanced representation.*

| Target | Group |
|---|---|
| 0 | A |
| 0 | B |
| 1 | C |
| 1 | A |
| 0 | A |

| Credibility Estimate |
|---|
| 3 * (0.33 – 0.4) |
| 1 * (0 – 0.4) |
| 1 * (0 – 0.4) |
| 3 * (0.33 – 0.4) |
| 3 * (0.33 – 0.4) |

*The more of a value we observe in a group, the more we trust that group's deviation from the overall mean.*

$$count_k \times (\bar{y}_k - \bar{y})$$

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# Feature Selection: Partial Dependence

**A Univariate Method:** The influence of each feature's influence is measured before modeling

Iteratively sets all observations in the column to each unique value contained in that column.

*Then*

Then observe the correlation each value of the column has to the response variable (target).

## Original Values

| Loan | Income | IsBad |
|---|---|---|
| $11,200 | $108,000 | False |
| $10,000 | $65,000 | False |
| $8,000 | $20,000 | True |
| $16,000 | $110,000 | True |
| $4,000 | $155,000 | False |

## All Values Set to $20k

| Loan | Income | IsBad |
|---|---|---|
| $11,200 | **$20,000** | False |
| $10,000 | **$20,000** | False |
| $8,000 | **$20,000** | True |
| $16,000 | **$20,000** | True |
| $4,000 | **$20,000** | False |

## All Values Set to $65k

| Loan | Income | IsBad |
|---|---|---|
| $11,200 | **$65,000** | False |
| $10,000 | **$65,000** | False |
| $8,000 | **$65,000** | True |
| $16,000 | **$65,000** | True |
| $4,000 | **$65,000** | False |

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# Feature Selection: Permutation Importance

**A Model-based Method:** The performance of the model is measured before and after...

Randomly shuffling the values in each column, one-at-a-time, to break the correlation that each column has to the target variable

*Then*

Measuring the impact the change to each column's influence has upon the model's overall performance according to one of many applicable metrics:

*Generates a stack-ranked list of features by their scores*

**Classification:** Accuracy, Precision, or Recall
**Regression:** MAE, RMSE, RAE, RSE, $R^2$

## Original Values

| Loan | Income | IsBad |
|---|---|---|
| $11,200 | $108,000 | False |
| $10,000 | $65,000 | False |
| $8,000 | $20,000 | True |
| $16,000 | $110,000 | True |
| $4,000 | $155,000 | False |

## Shuffle Column 1

| Loan | Income | IsBad |
|---|---|---|
| $8,000 | $108,000 | False |
| $4,000 | $65,000 | False |
| $16,000 | $20,000 | True |
| $10,000 | $110,000 | True |
| $11,200 | $155,000 | False |

## Shuffle Column 2

| Loan | Income | IsBad |
|---|---|---|
| $11,200 | **$65,000** | False |
| $10,000 | **$155,000** | False |
| $8,000 | **$108,000** | True |
| $16,000 | **$20,000** | True |
| $4,000 | **$110,000** | False |

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